

Report
Of
DATA SCIENCE MASTER VIRTUAL INTERNSHIP

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.
In partial fulfilment of the requirement of degree of
B.TECH CSE



By
Roll Number of Student: 2410030278
Name of Student: NAINIKA
Batch: 2024-2028

CANDIDATE’S DECLARATION

I, NAINIKA, do hereby declare that the internship report titled “Data Science Master Virtual Internship” has been completed by me in fulfilment of the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. This report is based on the learning activities, technical tracks and internship completion evidence associated with the Data Science Master Virtual Internship completed during April–June 2026. All references and external sources used in preparing this report have been properly acknowledged. I further declare that this report is my own academic work and has not been submitted to any other institute for any degree or diploma requirement. I shall be solely responsible for any copyright violation, if any, in this regard.

Signature

Student Name: NAINIKA

Roll No.: 2410030278

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to EduSkills, the National Internship Portal and all organisations associated with the **Data Science Master Virtual Internship** for providing a structured learning environment in data science. I am especially thankful for the opportunity to work through technical tracks covering data engineering, machine learning, applications and use cases, and platform administration. The internship helped me understand how data workflows progress from data access and preparation to model development, evaluation and deployment.

I am also grateful to the faculty of the School of Computer Science and Engineering, IILM University, Greater Noida, for their academic guidance and encouragement. Finally, I thank my parents and family for their constant support and motivation throughout the internship.

Signature

Student Name: NAINIKA

Roll No.: 2410030278

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3. INTERNSHIP COMPLETION CERTIFICATE



शिक्षा मंत्रालय
MINISTRY OF
EDUCATION



अखिल भारतीय तकनीकी शिक्षा परिषद्
All India Council for Technical Education



EduSkills®
Nation Building Through Skills



Certificate of Virtual Internship

This is to certify that

Nainika .

IILM University, Greater Noida

has successfully completed the 8-weeks

Data Science Master Virtual Internship

During April - June 2026

May this Internship learning propel you toward a bright and successful career.

Supported By

SIEMENS

G.V. Pai

Gaurav Pai
Head - Strategy and Operations
Siemens Industry Software (India) Pvt. Ltd

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Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education

[Signature]

Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills



Certificate ID : 467eb93a2fbe5f1a12c2

Student ID: STU699ae83c57aa71771759676



GRADE- O (Outstanding): 90-100 | E (Excellent): 80-89 | A (Very Good): 70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

Internship Offer Letter



Ref No: C16Y2026M091441280

Internship Offer Letter

Student Name : Nainika .
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering)
Institute Name : IILM University, Greater Noida
Internship Domain : Siemens Data Science Master
Internship Duration : April 2026 - June 2026
AICTE Student ID : STU699ae83c57aa71771759676

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

4. PROJECT DESCRIPTION

4.1 INTRODUCTION

This report provides a comprehensive overview of the 8 week Data Science Master Virtual Internship completed through the AICTE EduSkills Virtual Internship Program. The internship was conducted from April 2026 to June 2026 as part of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida. The main objective was to gain practical knowledge of data science and machine learning using the RapidMiner ecosystem and associated analytical tools.

The internship followed a systematic curriculum that combined theoretical concepts with hands-on activities, assessments and technical learning modules. It covered Data Engineering, Machine Learning, Applications and Use Cases, and Platform Administration. Major topics included data access, cleaning, transformation, text processing, macros, loops, Web APIs, classification, feature engineering, parameter optimisation, time series forecasting, and Python and R integration. The programme also covered the CRISP DM framework, deployment techniques, web applications, AI Hub administration, scheduling and real time scoring, offering an end to end view of the data science lifecycle. The overall experience enhanced practical capabilities and developed a better understanding of industry focused data science workflows.

4.2 ORGANIZATION PROFILE

EduSkills Academy is an ISO 9001:2015 and ISO/IEC 20000-1:2018 certified organisation working under the theme “Nation Building Through Skills.” EduSkills partners with the All India Council for Technical Education (AICTE) to run the AICTE–EduSkills Virtual Internship Program under the patronage of the AICTE National Internship Portal. For the Data Science Master Virtual Internship, the programme focused on building practical data-science and analytics capabilities through the RapidMiner (Altair) ecosystem, covering structured learning tracks in Data Engineering, Machine Learning, Applications & Use Cases, and Platform Administration to help students develop industry-oriented skills in data processing, machine learning, workflow development, deployment, and enterprise level analytics.

4.3 PROBLEM STATEMENT

Modern organisations handle large volumes of data obtained from files, databases, software applications, web services and distributed platforms. Converting this diverse data into accurate and meaningful decisions is challenging, as the data may be incomplete, inconsistent, unstructured or subject to frequent changes. Furthermore, developing a machine-learning model is only one part of the overall process; the model must also be evaluated, documented, reproduced, integrated with external tools, deployed and effectively managed during execution.

The primary problem addressed during the internship was therefore both academic and practical: developing a comprehensive understanding of the complete data-science lifecycle while gaining hands-on experience with tools and techniques that facilitate the transformation of raw data into reliable and deployable analytical solutions. The programme supported this objective through regular assessments and individual Professional and Master-level technical learning tracks.

4.4 PROJECT OBJECTIVES

- Develop practical understanding of data access, cleansing, transformation and preparation workflows.
- Study advanced data engineering concepts such as text processing, macros, loops and interaction with Web APIs.
- Understand classification, feature engineering, parameter optimisation and model evaluation.
- Explore time series forecasting and integration of R and Python models.
- Apply the CRISP-DM framework to structure a data science project from understanding through deployment.
- Understand deployment, web applications, AI Hub administration and real-time scoring.
- Connect individual learning modules into an end-to-end view of data engineering, machine learning and production-oriented analytics.

4.5 SCOPE OF THE PROJECT

The internship scope encompassed the entire analytical process, beginning with data collection and extending to concepts related to production deployment. Rather than concentrating on a single algorithm or dataset, it focused on building transferable technical skills relevant to business analytics, forecasting, operational processes, customer insights and predictive modelling.

The learning areas covered Data Engineering at Professional and Master levels; Machine Learning at Professional and Master levels; Applications & Use Cases at Professional and Master levels; and Platform Administration at the Master level. Collectively, these domains provided practical exposure to model development and the infrastructure supporting analytical workflows.

4.6 TECHNOLOGIES AND TOOLS USED

- Data Science Platform: Altair AI Studio / RapidMiner Ecosystem
- Data Engineering: Data Cleaning, Data Transformation, ETL Workflows
- Machine Learning: Classification, Feature Engineering, Parameter Optimisation
- Data Processing: Text Processing, Macros, Loops, Web APIs
- Programming & Integration: Python, R
- Data Analytics Framework: CRISP-DM
- Deployment & Administration: Altair AI Hub, Scheduling, Jobs, Queues, Real-Time Scoring

The internship covered major technologies and concepts including Altair AI Studio/RapidMiner, Altair AI Hub, Python, R, Web APIs, text processing, machine learning, data engineering, CRISP-DM, deployment, scheduling, and real-time scoring.

4.7 SYSTEM ARCHITECTURE

The overall architecture adopted during the internship for the Data Science Master Virtual Internship can be described through the following layers:

- **Data Layer:** Data is gathered and retrieved from files, databases and external services using various data access techniques.
- **Data Engineering Layer:** The obtained data is cleaned, transformed and organised using processing workflows, text operations, macros, loops and Web API interactions.
- **Machine Learning Layer:** Processed data is utilised for feature engineering, classification, parameter tuning and time series forecasting.
- **Integration Layer:** Python and R are incorporated into analytical workflows to perform specialised data science and statistical modelling tasks.
- **Application Layer:** The developed workflows are structured using the CRISP DM framework and implemented for practical data science applications.
- **Deployment Layer:** Analytical solutions are prepared for deployment using web applications, AI Hub execution, scheduling and deployment packages.
- **Administration Layer:** AI Hub and Platform Administration support projects, jobs, queues, execution environments, access management and real time scoring.

4.8 METHODOLOGY

The programme followed a progressive, assessment-driven methodology in which foundational workflow skills were followed by advanced processing and production-oriented concepts.

Step 1: Data Understanding and Access – Identify data sources and understand how structured and external data can be accessed.

Step 2: Data Preparation – Clean, transform and prepare information for reliable analytical use.

Step 3: Advanced Data Engineering – Apply text processing, macros, loops and Web API interactions to create flexible workflows.

Step 4: Model Development – Study classification and other machine-learning processes using prepared data.

Step 5: Feature Engineering and Optimisation – Improve model inputs and tune parameters systematically.

Step 6: Advanced Analytics – Explore time-series forecasting and integration with Python and R.

Step 7: Applications and Use Cases – Organise projects using CRISP-DM and consider evaluation and deployment.

Step 8: Platform Operations – Understand AI Hub administration, architecture, deployment packages and real-time scoring concepts.

4.8.1 DATA ENGINEERING

The Data Engineering tracks concentrated on preparing data for analysis.

The Professional level covered data access, cleansing and workflow development, while the Master-level activities progressed to text processing, macros, loops and Web API management. These areas highlight the importance of data engineering in analytics, since model performance relies strongly on data quality and workflow design. Macros and loops enable reusable process logic, text processing allows analysis of unstructured information beyond conventional tables, and Web APIs demonstrate how workflows can connect with external services and systems.

4.8.2 MACHINE LEARNING

The Machine Learning tracks focused on classification, feature engineering and parameter optimisation. Classification is used to predict categorical outcomes, feature engineering converts raw data into meaningful representations, and parameter optimisation helps identify appropriate settings for improving model performance.

The advanced learning component also included time-series forecasting and the integration of R and Python models. Forecasting is useful for analysing data where observations follow a temporal sequence, while R/Python integration enables specialised statistical methods and programming libraries to be incorporated into broader analytical workflows.

4.8.3 APPLICATIONS & USE CASES

This track concentrated on applying technical modelling skills to develop practical data-science solutions. A key component was the CRISP-DM framework, which organises projects into business understanding, data understanding, data preparation, modelling, evaluation and deployment.

The concepts of deployment and web applications highlighted an important principle: developing a model is not the final step of a project. Analytical value is achieved when results are successfully incorporated into a functional workflow, application or service that can be accessed and utilised by users or integrated with other systems.

4.8.4 PLATFORM ADMINISTRATION

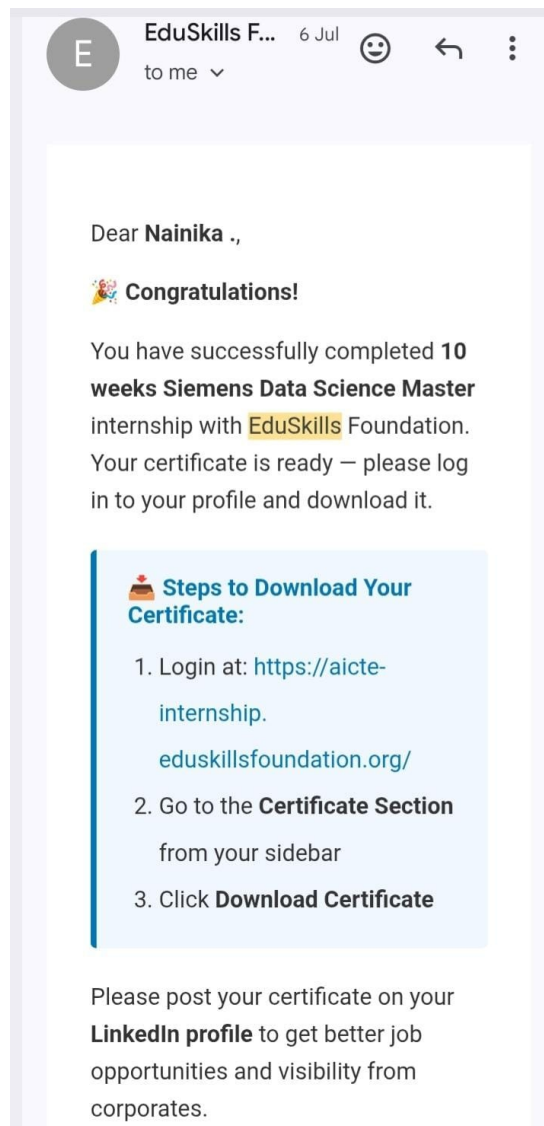
The Platform Administration Master component introduced production-oriented concepts around Altair AI Hub. Current Altair documentation describes AI Hub Server as managing projects, executions, queues, schedules and jobs, while Platform Admin supports administration of coding environments and Real Time Scoring deployments. Real-time scoring was particularly useful for understanding low-latency scenarios in which a process or model must respond quickly to requests. Platform administration also highlighted architecture, deployment packages, execution environments and access control, complementing model development with an operational perspective.

4.9 EXPECTED OUTCOMES

- Awareness of collaborative and administrative requirements involved in enterprise analytics.
- A stronger connection between academic machine-learning concepts and production-oriented data-science systems
- Ability to explain an end-to-end data workflow, from source data collection to final analytical results.
- Enhanced understanding of data cleaning, transformation and preparation techniques.
- Practical knowledge of text processing, macros, loops and Web API integration.
- Understanding of classification, feature engineering and parameter tuning methods.
- Familiarity with time series forecasting and integrated workflows using Python or R.
- Ability to apply CRISP-DM as a systematic framework for data-science projects.
- Understanding of deployment processes, web applications and real-time scoring.

4.10 CERTIFICATES OF COMPLETION AND OTHER COMMUNICATIONS

The Internship Completion Certificate issued for the successful completion of the Data Science Master Virtual Internship is attached in Chapter 3 of this report. The official internship completion email received from the concerned internship programme is also attached below as supporting communication proof. These documents serve as official evidence of the successful completion of the internship and the associated learning programme.



4.11 CONCLUSION

The Data Science Master Virtual Internship offered a systematic understanding of data science as a complete end-to-end process. The learning journey integrated data engineering, machine learning, applications and platform administration rather than approaching them as separate areas. Professional-level topics established essential skills in handling data and models, while Master-level concepts expanded this knowledge toward automation, integration and deployment.

The key outcome was a better understanding of how data workflows progress from data access and preparation through model development and controlled execution. Experience with advanced processing, CRISP-DM, Python/R integration, AI Hub administration and real-time scoring helped establish a valuable connection between academic data-science knowledge and practical, enterprise-focused analytical environments.

5. BIBLIOGRAPHY / REFERENCES

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